Anthill Antics

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October 30, 2026

1 Project Topic

My project will be on Langton's Ant, a cellular automata originally invented by Chris Langton. The core idea is that there's a grid of squares, on which an ant is walking. Depending on the color of the square, it'll either turn left or right, and then the square it just stepped on will change color.

2 Detailed Concept

First, let me talk about how this follows the idea of emergent behavior. The simulation starts with an incredibly simple trio of rules: When you step on a tile, switch its color from black to white or white to black. When you step on a black tile, turn right. When you step on a white tile, turn left. But this simple trio of rules leads to a chaotic dance as the ant spirals and loops around itself, drawing ever more complicated patterns and sometimes getting into long "highways" where it begins repeating the same path over and over.

Furthermore, more complicated rules can be given, with larger sequences of colors. There's even a few special patterns for colors which result in guaranteed symmetric patterns.

My design plan is to start with a fixed board size, and just two colors, like in the standard template. In addition, at first, I'd just be printing out each step of the simulation as text, since I don't yet have any experience with graphics. (Though, hopefully it'd be in a format that could be converted into a pretty-print somehow.)

Then, I'd like to make it possible for the user to set the size of the board by giving a height and width. In addition, I'd like to make it possible to set custom rules, up to a certain maximum length. (I'd set it up so that there's ten possible colors in a set order, and you can send in a rule like "RLLRRL" which turns into a set of instructions for which colors do what. If your rule isn't long enough for the whole set of colors, the system will only use as many colors as are needed to implement it.)

A really long-shot possibility would be to have multiple ants running concurrently, with a special rule for what happens if they collide (and maybe a special tile that splits the ants, too.) I don't know if that would be possible to implement, but it would be a really nice cherry on top.

Ultimately, I think I'd be satisfied if I could have

1. A board where the size can change according to user input.
2. The ability for a user to set up the board state before starting the ant.
3. At least one pattern other than the basic one, even if it has to be hardcoded rather than a user-input option.
4. The game can run on its own, and automatically stops if the ant reaches the border, or if the user presses a "stop" button.

It would also be nice if it could be paused and played, rather than 'stop' just halting the process completely and ending the simulation.

3 Expected Challenges

I already know how to set up the board, but figuring out how to efficiently move an ant around it might be a bit tricky. The things that I'm most worried about are the user-input portions, especially the part where I'm thinking about user-created rules. I have some ideas for how it could work, but they're all a bit tenuous at the moment.

I'm also really unsure how to do the "the game runs on its own until paused" thing, since that relies on time-based computation. I certainly *hope* there's a 'wait' or 'delay' function in Rust, or else the entire simulation will rush by in a fraction of a second due to how efficient compiled code is. It'd make pretty pictures, but it'd get rid of the whole point of watching the ant wander around the board. Making it a setting that the user can control would also be nice (as in, changing the time-per-step parameter either in a GUI or as part of the launch parameters).

I want the 'simulation' code to be entirely separate from the 'print' code, so that if I come up with a nice way of pretty-printing it with all its colors, I don't need to rewrite the game engine just to display it that way.